CO 2 ASSESSMENT TOOL 1-PART 2

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SMART NETWORK SECURITY USING REINFORCEMENT LEARNING

1) **TD (0)**
2) **SARSA**
3) **Q-LEARNING**

Adaptive Threat Detection and Response Strategy

1. Reinforcement Learning Environment

Component	**Network Security Meaning**	**Example**
State (S)	Current security condition	Normal / Suspicious / Intrusion
Action (A)	Available defensive decision	Monitor / Block / Isolate / Inspect
Reward (R)	Feedback from the decision	High for correct detection; penalty for unnecessary action
Policy (π)	Strategy learned by the agent	Best action for the current state
Value V(s)	Expected long-term usefulness of a state	How promising a security state is
Q-value Q (s, a)	Expected usefulness of an action in a state	Value of blocking a suspicious connection

2.Innovative workflow: OBSERVE → ASSESS → ACT → REWARD → LEARN → RESPOND. The same security state can lead to different actions depending on the learned value of future outcomes.

3. State and Action Design State Space:

S = {Normal Traffic, Suspicious Traffic, Intrusion Detected, Malware Activity, Recovery} Action Space:

A = {Monitor Traffic, Inspect Logs, Block Connection, Isolate Host, Generate Alert}

Eq. (1): $S = \{s_1, s_2, \ldots, s_n\}$

Eq. (2): $A = \{a_1, a_2, \ldots, a_m\}$

These definitions represent the possible security conditions and defensive actions available to the learning agent.

4. Intelligent Reward Design

A practical reward model should encourage correct detection while discouraging unnecessary or expensive responses.

Eq. (3): R = Detection Accuracy − Response Cost − False Alarm Penalty + Successful Mitigation Bonus

Interpretation:

- Correct threat identification → positive reward
- Successful containment → additional positive reward
- Unnecessary blocking → penalty
- Missed intrusion → strong penalty
- High computational/response cost → penalty

5. Core Reinforcement Learning Algorithms TD (0) — State Value Learning

TD (0) learns the value of a security state from the current experience and an estimate of the next state.

$$V(S_t) = V(S_t) + \alpha\,[R_{t+1} + \gamma V(S_{t+1}) - V(S_t)]$$

Pseudo Code:

Initialize V(s)=0 for every security state Repeat
for each security episode:
  Observe current state S_t
  Move to next state S_{t+1}
  Receive reward R_{t+1}
  Update $V(S_t)$
Until the values stabilize

SARSA — Safe On-Policy Learning

$$Q(S, A_t) = Q(S_t,A_t) + \alpha[R_{t+1} + \gamma Q(S_{t+1},A_{t+1}) - Q(S_t,A_t)]$$

SARSA updates using the action that the agent actually chooses next. This makes it an on-policy approach. **Pseudo Code:**

Initialize Q(s,a)
Choose A_t using ε-greedy policy Repeat:
  Execute A_t
  Observe R_{t+1} and S_{t+1}
  Choose A_{t+1}
  Update $Q(S_t,A_t)$
  Move to S_{t+1}, A_{t+1}

Until convergence

Q-Learning — Best-Future-Action Learning

$$Q(S_t, A_t) = Q(S_t, A_t) + \alpha[R_{t+1} + \gamma \max_a Q(S_{t+1},a) - Q(S_t,A_t)]$$

Q-Learning uses the maximum future Q-value and is therefore an off-policy approach.

Pseudo Code:

Initialize Q (s, a) Repeat:
   Observe current security state
   Select an action
   Execute action and receive reward
   Observe next state
   Update Q using max future Q-value
   Until convergence

6. TD(0) vs SARSA vs Q-Learning

Feature	**TD (0)**	**SARSA**	**Q-Learning**
Learns	State value	State-action value	State-action value
Policy	Value-based	On-policy	Off-policy
Update	Next-state value	Actual next action	Best next action
Main strength	Simple value estimation	Considers selected behavior	Targets best future action
Security use	Evaluate investigation states	Adaptive response policy	Optimal response selection

7. Mini Worked Example: Suspicious Network Event

Scenario: The agent observes a suspicious connection. It must choose between inspecting logs and blocking the connection.

Assume: $\alpha = 0.5$, $\gamma = 0.9$, current $Q(S_t, A_t) = 4$, reward = 8, and next-state Q-value = 6.

TD-style Q update using the selected next action:

New $Q = 4 + 0.5[8 + (0.9 \times 6) - 4]$

New $Q = 4 + 0.5[9.4] = 8.7$

The updated value becomes 8.7, showing that the observed security action was more valuable than previously estimated.

8. Decision Process

Step 1. Capture incoming network event or forensic evidence.

Step 2. Convert the event into the current security state.

Step 3. Read the learned Q-values for available actions.

Step 4. Select the action with the highest learned value.

Step 5. Execute the response and observe the result.

Step 6. Calculate the reward and update the learning model.

Step 7. Classify the event as normal, suspicious, intrusion, malware-related, or recovery-related.

Step 8. Store the decision and generate a security investigation report.

9. End-to-End Intelligent Security Flow

Network Event → State Identification → Action Selection → Security Response → Reward→ Q/V Update → Next Event → Adaptive Policy **10. Abbreviations & Symbols**

Symbol	**Meaning**
S	State Space
A	Action Space
R	Reward
V(s)	State Value Function
Q (s, a)	Action Value Function
α	Learning Rate
γ	Discount Factor
ε	Exploration Rate
t	Time Step
π	Policy / Decision Strategy